National Research University "Higher School of Economics".
Faculty of Computer Science. Data Science.
Ordered Sets in Data Analysis.
Homework №1 of Abramov Aleksandr Sergeevich.

Q1

Let R be the binary relation.

Property	***Does the property hold?***	***Explanation***
Reflexive	No	$(2; 2) \notin R$
Irreflexive	No	$(1; 1) \in R$
Symmetric	No	$(1; 4) \in R$ but $(4; 1) \notin R$
Asymmetric	No	$(1; 2) \in R$ and $(2; 1) \in R$
Antisymmetric	No	$(1; 2) \in R$ and $(2; 1) \in R$ but $1 \neq 2$
Transitive	No	$(2; 3) \in R$ and $(3; 6) \in R$ but $(2; 6) \notin R$
Linear	No	$(5; 7) \notin R$ and $(7; 5) \notin R$

Q2

Let R be the binary relation.

Property	***Does the property hold?***	***Explanation***
Reflexive	No	$(1; 1) \notin R$
Irreflexive	Yes	$\forall a\ (a; a) \notin R$
Symmetric	Yes	$\forall a, b\ aRb\ is\ false \Rightarrow (aRb \Rightarrow bRa)\ is\ true$
Asymmetric	Yes	$\forall a, b\ aRb\ is\ false \Rightarrow (aRb \Rightarrow \neg bRa)\ is\ true$
Antisymmetric	Yes	$\forall a, b\ aRb\ is\ false \Rightarrow (aRb\ \&\ bRa)\ is\ false \Rightarrow (aRb\ \&\ bRa \Rightarrow a = b)\ is\ true$
Transitive	Yes	$\forall a, b, c\ aRb\ is\ false \Rightarrow (aRb\ \&\ bRc)\ is\ false \Rightarrow (aRb\ \&\ bRc \Rightarrow aRc)\ is\ true$
Linear	No	$1 \neq 2$ but $(1, 2) \notin R$ and $(2, 1) \notin R$

<u>**Q3**</u>

$A = \{ "a", "ab", "abc", "ac", "b" \}, R = \{ (x, y) \mid x, y \in A \,\&\, x \, is \, a \, prefix \, of \, y \}$

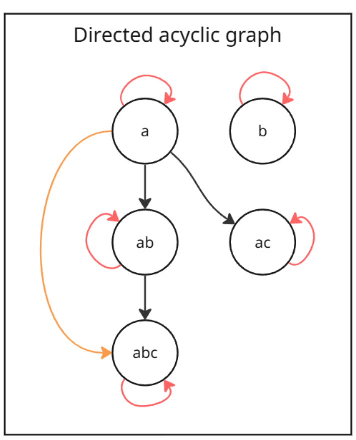
R	***a***	***ab***	***abc***	***ac***	***b***
a	1	1	1	1	0
ab	0	1	1	0	0
abc	0	0	1	0	0
ac	0	0	0	1	0
b	0	0	0	0	1

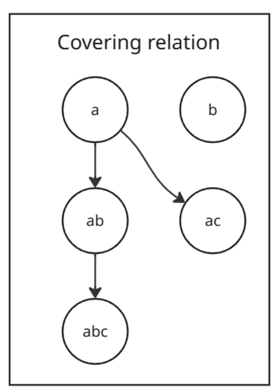
Prove that R is a partial order

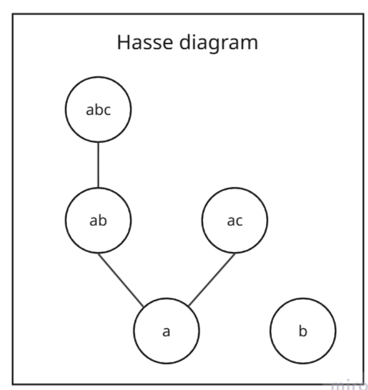
1. $\forall a \in A \ (a, a) \in R \Rightarrow R \, is \, reflexive$
2. $\forall a, b, c \in A \ aRb \,\&\, bRc \Rightarrow aRc$: if a is a prefix of b and b is a prefix of c than a is a prefix of c. In fact, only $a = "a", b = "ab", c = "abc"$ needs to be checked and the condition holds. Therefore, R is transitive
3. $\nexists \ a \neq b \ aRb \,\&\, bRa \Rightarrow R \, is \, antisymmetric$

R is reflexive, transitive and antisymmetric. Therefore, R is a partial order.

Draw its diagram (Hasse)

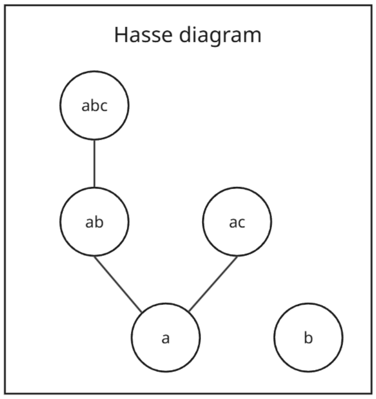


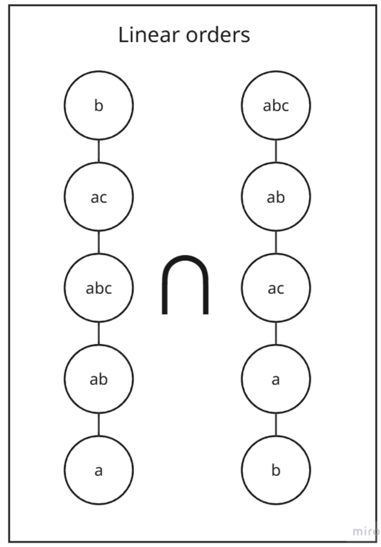




Determine the dimension of the partial order

1. The Hasse diagram is not a linear order. Therefore, the dimension is not 1.
2. The dimension is 2.





A = { (a, ab), (a, abc), (a, ac), (a, b), (ab, abc), (ab, ac), (ab, b), (abc, ac), (abc, b), (ac, b) }

B = { (b, a), (b, ac), (b, ab), (b, abc), (a, ac), (a, ab), (a, abc), (ac, ab), (ac, abc), (ab, abc) }

A ∩ *B* = { (a, ab), (a, abc), (a, ac), (ab, abc) }

Provide 3 different topological sortings

1. $\{\, a,\ ab,\ ac,\ abc,\ b \,\}$
2. $\{\, a,\ ab,\ abc,\ ac,\ b \,\}$
3. $\{\, b,\ a,\ ab,\ ac,\ abc \,\}$

Q4

Given:

$S_1 = \{ 1,\ 2,\ 3,\ 5,\ 6,\ 10,\ 15,\ 30 \},\ R_1 = (S_1,\ |)$

$S_2 = \{ \emptyset,\ x,\ y,\ z,\ xy,\ yz,\ xz,\ xyz \},\ R_2 = (S_2,\ \subseteq)$

Prove: $R_1 \cong R_2$

1) *Prove that R_1 is a partial order.*

   a) R_1 is reflexive: $\forall a \in S_1\ a \mid a$

   b) R_1 is transitive.

Proof:

$$\forall a, b, c \in S_1\ such\ that\ a|b \,\&\, b|c \Rightarrow \exists k \in \mathbb{N}:\ b = k \cdot a,\ \exists m \in \mathbb{N}:\ c = m \cdot b \Rightarrow$$
$$c = m \cdot (k \cdot a) = (m \cdot k) \cdot a = t \cdot a \Rightarrow \exists t \in \mathbb{N}: c = t \cdot a \Rightarrow a|c$$

   c) R_1 is antisymmetric.

Proof:

Assume the opposite. Then, $\exists a \neq b \in S_1\ a|b \,\&\, b|a \Rightarrow$
$$\exists k \in \mathbb{N},\ k \neq 1:\ b = k \cdot a,\ \exists m \in \mathbb{N},\ m \neq 1:\ a = m \cdot b \Rightarrow$$
$$a = m \cdot (k \cdot a) = (m \cdot k) \cdot a \Rightarrow m \cdot k = 1.$$
However, this may not hold with $m,\ k \in \mathbb{N},\ m, k \neq 1$. By contradiction, R_1 is antisymmetric.

   d) R_1 is reflexive, transitive, and antisymmetric. By definition, R_1 is a partial order.

2) *Prove that R_2 is a partial order.*

   a) R_2 is reflexive: $\forall a \in S_2\ a \subseteq a$

   b) R_2 is transitive.

Proof:

$$\forall a, b, c \in S_2\ such\ that\ a \subseteq b \,\&\, b \subseteq c \Rightarrow (1)\ \forall x \in a:\ x \in b,\ (2)\ \forall y \in b:\ y \in c \Rightarrow$$
$\forall x \in a,\ x \in b\ from\ (1) \Rightarrow x \in c\ from\ (2)$. Thus, $\forall x \in a,\ x \in c \Rightarrow a \subseteq c$

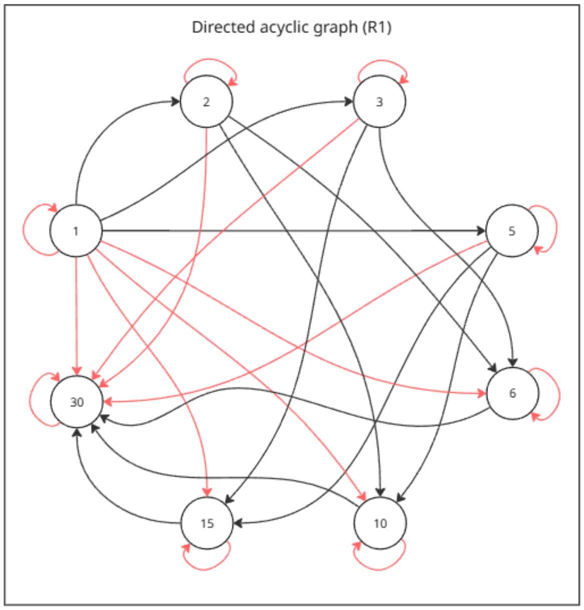
   c) R_2 is antisymmetric.

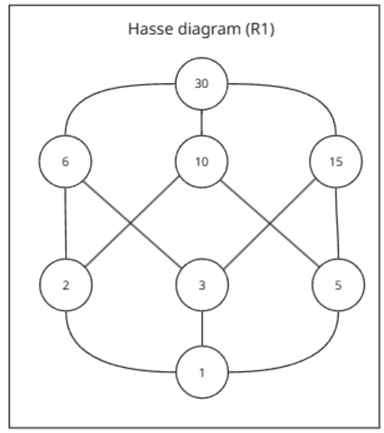
Proof:

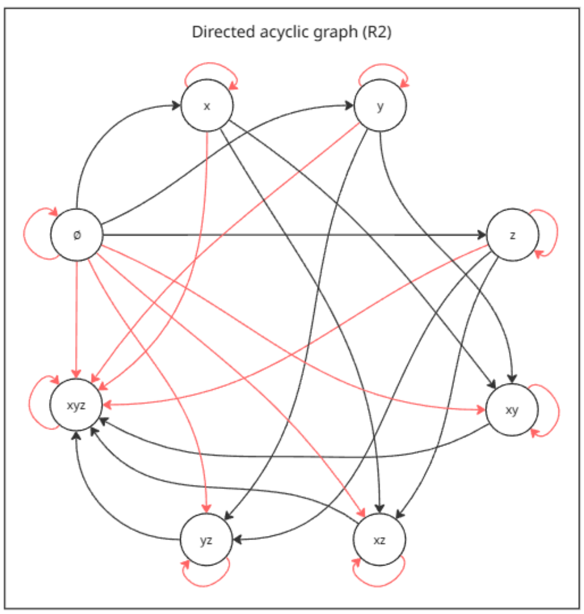
Assume the opposite. Then, $\exists a \neq b \in S_2\ a \subseteq b \,\&\, b \subseteq a \Rightarrow$
$(1)\ \forall x \in a:\ x \in b,\ (2)\ \forall x \in b:\ x \in a$. From $a \neq b$, $(3)\ \exists y \in a,\ y \notin b$ or $(4)\ \exists y \in b,\ y \notin a$.
However, (3) contradicts (1) and (4) contradicts (2). Therefore, neither (3) nor (4) is true, and $a \neq b$ is false. Thus, $a = b$ and R_2 is antisymmetric.

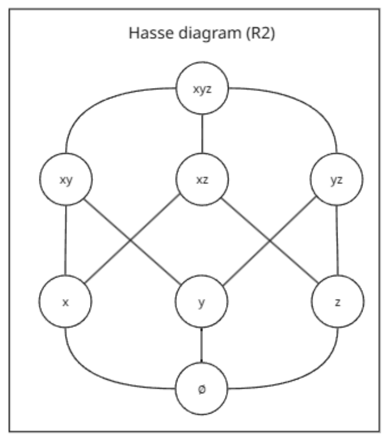
   d) R_2 is reflexive, transitive, and antisymmetric. By definition, R_2 is a partial order.

3) Construct Hasse diagrams of R_1 and R_2.









Hasse diagrams of R_1 and R_2 are similar. Therefore, R_1 and R_2 are isomorphic. Explicitly, the isomorphism can be described as follows:

$\emptyset \leftrightarrow 1$
$x \leftrightarrow 2$
$y \leftrightarrow 3$
$z \leftrightarrow 5$
$xy \leftrightarrow 6$
$xz \leftrightarrow 10$
$yz \leftrightarrow 15$
$xyz \leftrightarrow 30$

Q5

$S = \{1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60\}$, $R = (S, |)$

Prove that R is a partial order

1) R is reflexive: $\forall a \in S\ a \mid a$
2) R is transitive.

> Proof:
> $$\forall a, b, c \in S \text{ such that } a|b \,\&\, b|c \Rightarrow \exists k \in \mathbb{N}:\ b = k \cdot a,\ \exists m \in \mathbb{N}:\ c = m \cdot b \Rightarrow$$
> $$c = m \cdot (k \cdot a) = (m \cdot k) \cdot a = t \cdot a \Rightarrow \exists t \in \mathbb{N}: c = t \cdot a \Rightarrow a|c$$

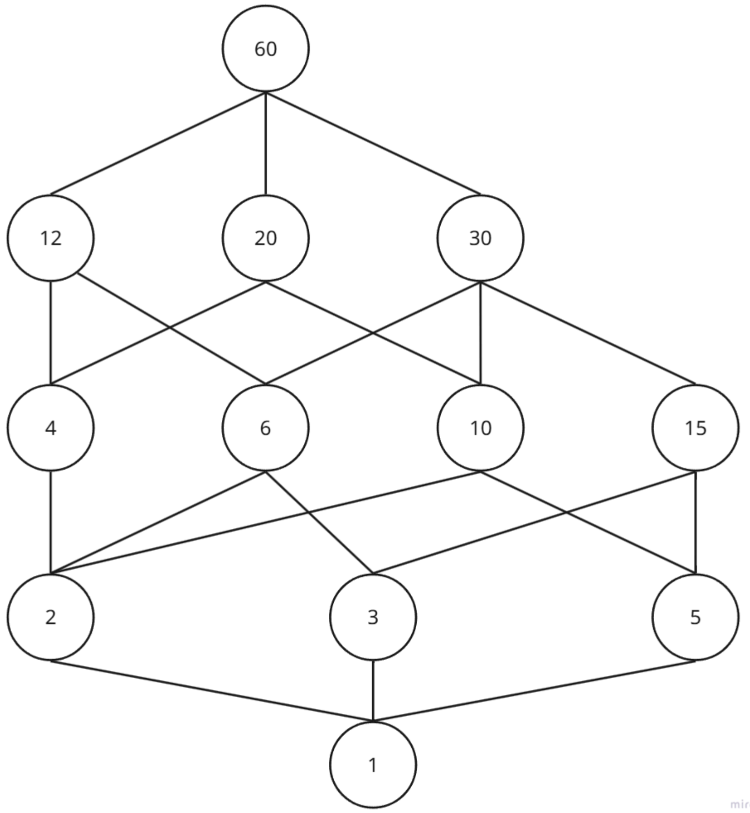
3) R is antisymmetric.

> Proof:
> Assume the opposite. Then, $\exists a \neq b \in S\ a|b \,\&\, b|a \Rightarrow$
> $\exists k \in \mathbb{N},\ k \neq 1:\ b = k \cdot a,\ \exists m \in \mathbb{N},\ m \neq 1:\ a = m \cdot b \Rightarrow$
> $a = m \cdot (k \cdot a) = (m \cdot k) \cdot a \Rightarrow m \cdot k = 1.$
> However, this may not hold with $m,\ k \in \mathbb{N},\ m, k \neq 1$. By contradiction, R is antisymmetric.

4) R is reflexive, transitive, and antisymmetric. By definition, R is a partial order.

Draw the diagram of R

R	**1**	**2**	**3**	**4**	**5**	**6**	**10**	**12**	**15**	**20**	**30**	**60**
1	1	1	1	1	1	1	1	1	1	1	1	1
2		1		1		1	1	1		1	1	1
3			1			1		1	1		1	1
4				1				1		1		1
5					1		1		1	1	1	1
6						1		1			1	1
10							1			1	1	1
12								1				1
15									1		1	1
20										1		1
30											1	1
60												1



Answers

1. Maximal elements – { 60 }
2. Largest element – 60
3. Minimal elements – { 1 }
4. Least element – 1
5. $\downarrow \{6,\ 15\} = \{6,\ 15,\ 2,\ 3,\ 5,\ 1\}$
6. $\uparrow \{6,\ 15\} = \{6,\ 15,\ 12,\ 30,\ 60\}$
7. $\downarrow \{4,\ 10\} = \{4,\ 10,\ 2,\ 5,\ 1\}$
8. $\uparrow \{4,\ 10\} = \{4,\ 10,\ 12,\ 20,\ 30,\ 60\}$
9. $\downarrow \{12,\ 30\} = \{12,\ 30,\ 4,\ 6,\ 10,\ 15,\ 2,\ 3,\ 5,\ 1\}$
10. $\uparrow \{12,\ 30\} = \{12,\ 30,\ 60\}$

Q6

1. $f \wedge m = f$
2. $a \vee l = l$
3. $i \wedge k = a$
4. $a \wedge (c \vee d) = a \wedge l = a$
5. $(i \wedge g) \vee (c \wedge d) = g \vee a = g$

6. $\vee\ (b,\ c,\ d) = l$
7. $\wedge\ \emptyset = 1$
8. $\vee\ \emptyset = 0$
9. Upper semi-lattice: true
10. Lower semi-lattice: true
11. Lattice: true

Q7

1. Prove that $\forall x, y, z \in L\ x \wedge y \leq x \vee z$
   1.1. By definition of infimum, $x \wedge y \leq x$
   1.2. By definition of supremum, $x \leq x \vee z$
   1.3. From (1.1) and (1.2), $x \wedge y \leq x \leq x \vee z \Rightarrow x \wedge y \leq x \vee z$
2. Prove that $\forall a, b, c, d \in L\ (a \wedge c) \leq (a \vee b) \wedge (c \vee d)$
   2.1. $a \wedge c \leq a \vee b$ – from (1) with $x = a,\ y = c,\ z = b$
   2.2. $c \wedge a \leq c \vee d$ – from (1) with $x = c,\ y = a,\ z = d$
   2.3. By definition, $a \wedge c = c \wedge a$
   2.4. From (2.1) – (2.3), $(a \wedge c) \in LB\{\ a \vee b,\ c \vee d\ \}$
   2.5. By definition of infimum and lower bound, $(a \wedge c) \leq (a \vee b) \wedge (c \vee d)$
3. Similarly, $\forall a, b, c, d \in L\ (b \wedge d) \leq (a \vee b) \wedge (c \vee d)$
   3.1. $b \wedge d \leq b \vee a$ – from (1) with $x = b,\ y = d,\ z = a$
   3.2. $d \wedge b \leq d \vee c$ – from (1) with $x = d,\ y = b,\ z = c$
   3.3. By definition, $b \wedge d = d \wedge b, b \vee a = a \vee b, d \vee c = c \vee d$
   3.4. From (3.1) – (3.3), $(b \wedge d) \in LB\{\ a \vee b,\ c \vee d\ \}$
   3.5. By definition of infimum and lower bound, $(b \wedge d) \leq (a \vee b) \wedge (c \vee d)$
4. From (2) and (3), $(a \vee b) \wedge (\ c \vee d\ ) \in UB\{\ a \wedge c,\ b \wedge d\ \}$
5. By definition of supremum and upper bound, $(a \vee b) \wedge (\ c \vee d\ ) \geq (a \wedge c) \vee (b \wedge d)$. Q.E.D.

Q8

1. Left lattice
   1.1. Not modular: contains a pentagon $\{\ 1,\ a,\ c,\ 0,\ z\ \}$. Check: $c \leq a \rightarrow c \vee (z \wedge a) = (c \vee z) \wedge a$
      1.1.1. $c \leq a$ is true
      1.1.2. $c \vee (z \wedge a) = c \vee 0 = c$
      1.1.3. $(c \vee z) \wedge a = 1 \wedge a = a$
      1.1.4. $c \neq a \Rightarrow$ not modular by definition
   1.2. Not distributive because not modular. Actually, let's explicitly find a diamond $\{\ 1,\ c,\ y,\ x,\ 0\ \}$. Check $c \wedge (y \vee x) = (c \wedge y) \vee (c \wedge x)$

1.2.1. $c \wedge (y \vee x) = c \wedge 1 = c$

1.2.2. $(c \wedge y) \vee (c \wedge x) = 0 \wedge 0 = 0$

1.2.3. $c \neq 0 \Rightarrow$ not distributive by definition

2. Right lattice: contains a pentagon $\{g, e, b, f, a\}$
   2.1. Not modular: check $b \leq e \rightarrow b \vee (f \wedge e) = (b \vee f) \wedge e$
      2.1.1. $b \leq e$ is true
      2.1.2. $b \vee (f \wedge e) = b \vee a = b$
      2.1.3. $(b \vee f) \wedge e = g \wedge e = e$
      2.1.4. $b \neq e \Rightarrow$ not modular by definition
   2.2. Not distributive since not modular. Let us explicitly check $e \wedge (b \vee f) = (e \wedge b) \vee (e \wedge f)$
      2.2.1. $e \wedge (b \vee f) = e \wedge g = e$
      2.2.2. $(e \wedge b) \vee (e \wedge f) = b \vee a = b$
      2.2.3. $e \neq b \Rightarrow$ not distributive by definition

Q9

1.

$$\text{③}\;\dfrac{\text{Given: } X \rightarrow Z \qquad \dfrac{.}{Z \cup X \rightarrow Z \cup X}\;\text{①}}{X \cup X \rightarrow Z \cup X}$$

$$X \rightarrow Z \cup X \quad \text{Q.E.D.}$$

2.

$$\dfrac{\text{Given: } X \rightarrow W \qquad \begin{array}{c} \text{③}\;\dfrac{\text{Given: } X \rightarrow Z \qquad \dfrac{.}{Z \cup W \rightarrow Z \cup W}\;\text{①}}{X \cup W \rightarrow Z \cup W} \\ W \cup X \rightarrow Z \cup W \end{array}}{X \cup X \rightarrow Z \cup W}\;\text{③}$$

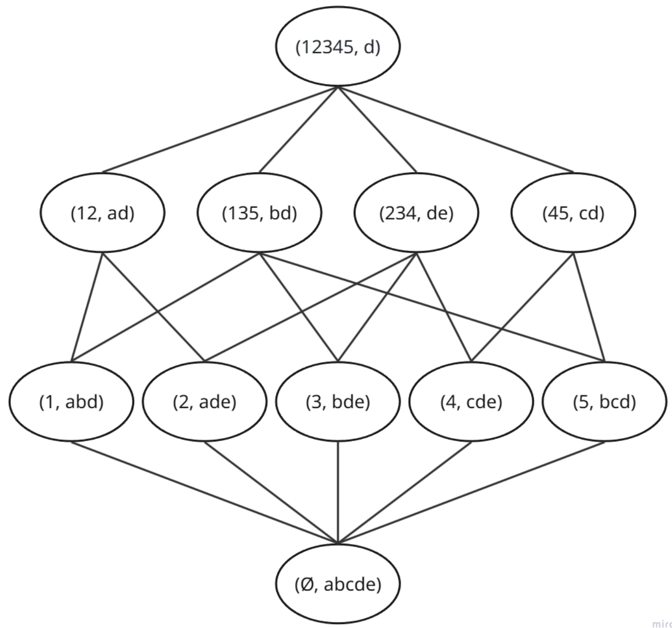
$$X \rightarrow Z \cup W \quad \text{Q.E.D.}$$

Q10

	FC?	MG?	PI?	PP?		FC?	MG?	PI?	PP?		FC?	MG?	PI?	PP?
$\varnothing'' = G' = d$		+	+	+	$bd'' = 135' = bd$	+				$bcd'' = 5' = bcd$	+			
$a'' = 12' = ad$		+	-	-	$be'' = 3' = bde$		+	-	-	$bce'' = \varnothing' = M$		+	-	+
$b'' = 135' = bd$		+	-	-	$cd'' = 45' = cd$	+				$bde'' = 3' = bde$	+			
$c'' = 45' = cd$		+	-	-	$ce'' = 4' = cde$		+	-	-	$cde'' = 4' = cde$	+			
$d'' = 12345' = d$	+				$de'' = 234' = de$	+				$abcd'' = \varnothing' = M$		-	-	-
$e'' = 234' = de$		+	-	-	$abc'' = \varnothing' = M$		-	-	-	$abce'' = \varnothing' = M$		-	-	-
$ab'' = 1' = abd$		+	-	-	$abd'' = 1' = abd$	+				$abde'' = \varnothing' = M$		-	+	-
$ac'' = \varnothing' = M$		+	-	+	$abe'' = \varnothing' = M$		+	-	+	$bcde'' = \varnothing' = M$		-	+	-
$ad'' = 12' = ad$	+				$acd'' = \varnothing' = M$		-	+	-	$acde'' = \varnothing' = M$		-	-	-
$ae'' = 2' = ade$		+	-	-	$ace'' = \varnothing' = M$		-	-	-	$M'' = \varnothing' = M$	+			
$bc'' = 5' = bcd$		+	-	-	$ade'' = 2' = ade$	+								

* FC – formal concept
** MG – minimal generator
*** PI – pseudo-intent
**** PP – proper premise

1. Formal concepts: $(12345,\ d)$, $(12,\ ad)$, $(135,\ bd)$, $(45,\ cd)$, $(234,\ de)$, $(1,\ abd)$, $(2,\ ade)$, $(5,\ bcd)$, $(3,\ bde)$, $(4,\ cde)$, $(\varnothing,\ abcde)$
2. Concept lattice diagram:



3.

Closed subsets	Non-trivial minimal generators	Generator implication cover
d	$\emptyset$	$\emptyset \to d$
ad	a	$a \to d$
bd	b	$b \to d$
cd	c	$c \to d$
de	e	$e \to d$
abd	ab	$ab \to d$

ade	ae	$ae \to d$
bcd	bc	$bc \to d$
bde	be	$be \to d$
cde	ce	$ce \to d$
$abcde$	ac, abe, bce	$ac \to bde,\ abe \to cd,\ bce \to ad$

4. Minimum basis of implication: $\emptyset \to d,\ acd \to be,\ abde \to c,\ bcde \to a$
5. Direct basis: $\emptyset \to d,\ ac \to bde,\ abe \to cd,\ bce \to ad$